

- Description

This module manages a vehicle speed monitoring system, ensuring that:

1. Each vehicle's speed is recorded and checked against the speed limit.
2. If a vehicle exceeds the speed limit, a fine is issued based on the severity of the violation.
3. Speed violations are categorized into minor, moderate, and severe, with increasing fine amounts.
4. A log of all recorded speeds, violations, and fines is maintained for law enforcement tracking.

This system ensures road safety by enforcing speed limits and penalizing excessive speeding.

- Specification

module SpeedMonitoringSystem

var speed, speedLimit, fineAmount: real

var violationLog: sequence of string

process checkSpeed(licensePlate: string, speed: real) fineAmount: real

functional scenario form:

speed <= speedLimit && fineAmount = 0, No violation.

||

speed - speedLimit <= 10 && fineAmount = 50 and violationLog = violationLog

⊕ ("Minor violation: " + licensePlate), Minor violation, fine: \$50.

||

speed - speedLimit <= 30 && fineAmount = 200 and violationLog =

violationLog ⊕ ("Moderate violation: " + licensePlate), Moderate violation, fine: \$200.

||

speed - speedLimit > 30 && fineAmount = 500 and violationLog = violationLog

⊕ ("Severe violation: " + licensePlate), Severe violation, fine: \$500.

end process

end module

- Java Code

```
package code;
import java.util.*;
class SpeedViolation {
    private String licensePlate;
    private double speed;
    private double speedLimit;
    private double fineAmount;
    public SpeedViolation(String licensePlate, double speed, double speedLimit) {
        this.licensePlate = licensePlate;
```

```

        this.speed = speed;
        this.speedLimit = speedLimit;
        this.fineAmount = calculateFine();
    }
    private double calculateFine() {
        double overSpeed = speed - speedLimit;
        if (overSpeed <= 0) {
            return 0.0;
        } else if (overSpeed <= 10) {
            return 50.0;
        } else if (overSpeed <= 30) {
            return 200.0;
        } else {
            return 500.0;
        }
    }
    public boolean isViolation() {
        return fineAmount > 0;
    }
    public String getViolationDetails() {
        if (isViolation()) {
            return String.format("License Plate: %s | Speed: %.1f km/h  
| Limit: %.1f km/h | Fine: $%.2f",
                licensePlate, speed, speedLimit, fineAmount);
        } else {
            return String.format("License Plate: %s | Speed: %.1f km/h  
| Limit: %.1f km/h | No Violation",
                licensePlate, speed, speedLimit);
        }
    }
}

class SpeedMonitor {
    private double speedLimit;
    private List<String> violationLog;
    public SpeedMonitor(double speedLimit) {
        this.speedLimit = speedLimit;
        this.violationLog = new ArrayList<>();
    }
    public void recordSpeed(String licensePlate, double speed) {
        SpeedViolation violation = new SpeedViolation(licensePlate, speed, speedLimit);
        System.out.println(violation.getViolationDetails());
        if (violation.isViolation()) {
            violationLog.add(violation.getViolationDetails());
        }
    }
}

```

```

    }
}
public void printViolationLog() {
    System.out.println("\nSpeed Violation Log:");
    for (String log : violationLog) {
        System.out.println(log);
    }
}
}

public class c18_VehicleSpeedSystem {
    public static void main(String[] args) {
        SpeedMonitor monitor = new SpeedMonitor(60.0);
        // testcase-VT:
        //     monitor.recordSpeed("ABC123", 55.0);
        //     monitor.recordSpeed("XYZ789", 65.0);
        //     monitor.recordSpeed("LMN456", 85.0);
        //     monitor.recordSpeed("QWE222", 120.0);

        // testcase-FT:
        //     monitor.recordSpeed("", -1.7976931348623157E308);
        //     monitor.recordSpeed("", 0);
        //     monitor.recordSpeed("4USS", -4.9E-324);
        //     monitor.recordSpeed("???", -9.658194489260283E307);

        // testcase-Z3:
        //     monitor.recordSpeed("ABC123", 0);
        //     monitor.recordSpeed("XYZ789", -1);
        //     monitor.recordSpeed("LMN456", 31);
        //     monitor.recordSpeed("QWE222", 9);

        // testcase-VT:
        monitor.recordSpeed("ABC123", 55);
        monitor.recordSpeed("XYZ789", 65);
        monitor.recordSpeed("LMN456", 85);
        monitor.recordSpeed("QWE222", 12);

        monitor.printViolationLog();
    }
}

```

- Coverage report

1. Test case number: 4
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.7 %	14	828	842
▼ src	1.7 %	14	828	842
▼ code	1.7 %	14	828	842
▼ c18_VehicleSpeedSystem.java	100.0 %	14	0	14
> c18_VehicleSpeedSystem		0	0	0
> SpeedMonitor	100.0 %	4	0	4
> SpeedViolation	100.0 %	10	0	10

3. Vibration testing instruction coverage: 98.4%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	1.1 %	180	16,428	16,608
▼ src	1.1 %	180	16,428	16,608
▼ code	1.1 %	180	16,428	16,608
▼ c18_VehicleSpeedSystem.java	98.4 %	180	3	183
> c18_VehicleSpeedSystem	88.9 %	24	3	27
> SpeedMonitor	100.0 %	52	0	52
> SpeedViolation	100.0 %	104	0	104

4. Fuzz testing branch coverage: 35.7%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.6 %	5	837	842
▼ src	0.6 %	5	837	842
▼ code	0.6 %	5	837	842
▼ c18_VehicleSpeedSystem.java	35.7 %	5	9	14
> SpeedViolation	30.0 %	3	7	10
> SpeedMonitor	50.0 %	2	2	4
> c18_VehicleSpeedSystem		0	0	0

5. Fuzz testing instruction coverage: 67.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.7 %	123	16,485	16,608
▼ src	0.7 %	123	16,485	16,608
▼ code	0.7 %	123	16,485	16,608
▼ c18_VehicleSpeedSystem.java	67.2 %	123	60	183
> SpeedViolation	57.7 %	60	44	104
> SpeedMonitor	75.0 %	39	13	52
> c18_VehicleSpeedSystem	88.9 %	24	3	27

6. Z3 branch coverage: 35.7%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.6 %	5	835	840
▼ src	0.6 %	5	835	840
▼ code	0.6 %	5	835	840
▼ c18_VehicleSpeedSystem.java	35.7 %	5	9	14
> SpeedViolation	30.0 %	3	7	10
> SpeedMonitor	50.0 %	2	2	4
> c18_VehicleSpeedSystem		0	0	0

7. Z3 instruction coverage: 67.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	123	21,972	22,095
▼ src	0.6 %	123	21,972	22,095
▼ code	0.6 %	123	21,972	22,095
▼ c18_VehicleSpeedSystem.java	67.2 %	123	60	183
> SpeedViolation	57.7 %	60	44	104
> SpeedMonitor	75.0 %	39	13	52
> c18_VehicleSpeedSystem	88.9 %	24	3	27

8. Unimproved V-method branch coverage: 92.9%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.5 %	13	827	840
▼ src	1.5 %	13	827	840
▼ code	1.5 %	13	827	840
▼ c18_VehicleSpeedSystem.java	92.9 %	13	1	14
> SpeedViolation	90.0 %	9	1	10
> SpeedMonitor	100.0 %	4	0	4
> c18_VehicleSpeedSystem		0	0	0

9. Unimproved V-method instruction coverage: 97.3%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼  VibrationTestingTool	 0.8 %	178	21,917	22,095
▼  src	 0.8 %	178	21,917	22,095
▼  code	 0.8 %	178	21,917	22,095
▼  c18_VehicleSpeedSystem.java	 97.3 %	178	5	183
>  c18_VehicleSpeedSystem	 88.9 %	24	3	27
>  SpeedViolation	 98.1 %	102	2	104
>  SpeedMonitor	 100.0 %	52	0	52